

This Booklet contains printed pages = 16

Booklet No.: 27274

22 (PGEE) MCA

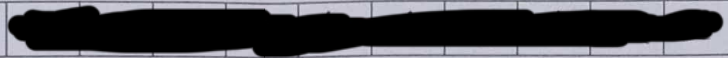
Test Booklet Code

CPGEE-2023
MCA



Time : One Hour Thirty Minutes

APPLICATION ID



INSTRUCTIONS

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO.

Read the following instructions carefully before you start answering the questions:

1. Fill in the particulars **Application ID**, **Test Booklet Code** on the Answer Sheet (OMR sheet) carefully with **blue/black** ball point pen only. The OMR Sheet contains a Carbon Copy which should not be detached from the Original Copy while writing particulars and marking responses on the Original Copy.
2. Please ensure that the Test Booklet is properly printed with all questions immediately after opening the Booklet. In case any discrepancy is detected, please report the same to the Invigilator immediately before answering.
3. If any damage or discrepancy is detected on the part of OMR sheet, it should be reported to the Invigilator immediately before answering. If the candidate starts writing on the OMR, it will not be changed.
4. The test is of **1 hour 30 minutes** duration and the Test Booklet contains 60 multiple choice questions each having **Four** options with **a single correct answer**.
5. Each question carries **4 (four)** marks. For each correct response, **4 (four)** marks will be awarded and for each incorrect response, **1(One)** mark will be deducted. There will be no marking for unattempted question.
6. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate **must detach the Carbon Copy from the OMR Sheet and hand over the ORIGINAL OMR sheet to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away the Carbon Copy and this Test Booklet with them.
8. The candidates should ensure that **OMR sheet** is not folded. Do not make any stray marks on the OMR. Do not write your Application ID anywhere else except the specified space in this Booklet and OMR Sheet.
9. Use of eraser, blade, chemical whitener fluid to rectify any response is **NOT** permissible on the OMR Sheet.
10. No candidate shall be admitted to the Examination Hall / Room 15 minutes after the commencement of the examination.
11. No candidate shall leave the Examination Hall / Room without prior permission of the Invigilator. No candidate shall be permitted to hand over his/her Answer Sheet and leave the Examination Hall / Room before expiry of the full time allotted for the paper.
12. The candidate shall not have in his /her possession inside the Examination Hall any Mobile, Electronic gadgets, calculator, book, notebook, loose paper in any form etc. except Admit Card, required ball pens and water bottle.
13. Any violation of the examination rules/discipline on the part of a candidate, he/she shall be expelled from the examination.

SE

Signature of Candidate

Signature of the Invigilator

There are total 60 questions in this Test Booklet. Each question is having four alternative responses, of which one response is completely correct. Record the completely correct response on the OMR sheet by darkening the corresponding circle with Blue/Black ball point pen only.

1. New data are to be inserted into a stack data structure, but there is no available space. This situation is usually called as
(A) underflow
(B) overflow
(C) houseful
(D) saturated
2. What is information about data called?
(A) Hyper data
(B) Tera data
(C) Meta data
(D) Relations
3. Which of the following is not a type of database?
(A) Hierarchical
(B) Network
(C) Distributed
(D) Decentralized
4. In SQL, which command is used to add new rows to a table?
(A) ALTER TABLE
(B) ADD ROW
(C) INSERT INTO
(D) APPEND
5. A functional dependency of the form $X \rightarrow Y$ is trivial if
(A) $Y \subseteq X$
(B) $Y \subset X$
(C) $X \subseteq Y$
(D) $X \subset Y$ and $Y \subset X$
6. Which of the following is TRUE?
(A) Every relation in 3NF is also in BCNF.
(B) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R.
(C) Every relation in BCNF is also in 3NF.
(D) No relation can be in both BCNF and 3NF.

7. Characters are grouped into tokens in which of the following phase of the compiler design?
(A) Code generator (B) Lexical analyzer
(C) Parser (D) Code optimization
8. The context free grammar is ambiguous if _____
(A) The grammar contains non-terminals
(B) Produces more than one parse tree ✓
(C) Production has two non-terminals side by side
(D) The grammar contains only terminals
9. Let the class of language accepted by finite state machine be L_1 and the class of language represented by regular expressions be L_2 then?
(A) $L_1 < L_2$ (B) $L_1 \geq L_2$
(C) $L_1 \cup L_2 = \Sigma^*$ (D) $L_1 = L_2$
10. A Push Down Automata is _____, if there is at most one configuration.
(A) Deterministic (B) Non deterministic
(C) Finite (D) Non finite
11. P, Q, R be regular expression over Σ , P is not ϵ , then $R = Q + RP$ has a unique solution:
(A) Q^*P (B) QP^*
(C) Q^*P^* (D) $(P^*Q)^*$
12. RR^* can be expressed in which of the forms:
(A) R^+ (B) R^-
(C) $R^+ \cup R^-$ (D) R
13. The regular expression $(a + b)^*$ denotes a language that consists of:
(A) Strings containing only the letter "a".
(B) Strings containing only the letter "b".
(C) Strings containing any combination of "a" and "b". ✓
(D) Strings containing either "a" or "b", but not both.

14. Suppose that a process is in 'BLOCKED' state waiting for some I/O service. When the service is completed, it goes to the

(A) RUNNING state

(B) READY state

(C) SUSPENDED state

(D) TERMINATED state

15. Let the time taken to switch between user and kernel modes of execution be t_1 while the time taken to switch between two processes be t_2 . Which of the following is TRUE?

(A) $t_1 > t_2$

(B) $t_1 = t_2$

(C) $t_1 < t_2$

(D) Nothing can be said about the relation between t_1 and t_2 .

Switching between user and kernel mode is faster than switching between two processes.

16. What will be the output of the following program?

```
int main()
{
    int i;
    for (i = 0; i < 10; ++i)
        printf ("%d", i & 1);
}
```

i (in binary)	i & 1	result	value
0 (0000)	0000 & 0001	0000	0 = even
1 (0001)	0001 & 0001	0001	1 = odd
2 (0010)	0010 & 0001	0000	0 = even
3 (0011)	0011 & 0001	0001	1 = odd
4 (0100)	0100 & 0001	0000	0 = even
5 (0101)	0101 & 0001	0001	1 = odd
6 (0110)	0110 & 0001	0000	0 = even
7 (0111)	0111 & 0001	0001	1 = odd
8 (1000)	1000 & 0001	0000	0 = even
9 (1001)	1001 & 0001	0001	1 = odd

(A) 0101010101

(B) 0000000000

(C) 1111111111

(D) 0000011111

17. What will be the output of the following program?

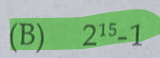
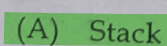
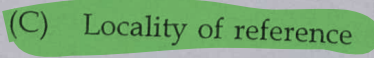
```
int main ()
{
    int i = 67, x=5;
    printf((x<4)? "%d": "%c", i);
}
```

(A) c

(B) C

(C) A

(D) 67

18. If an integer needs two bytes of storage, then maximum value of a signed integer is
(A) $2^{16}-1$  (B) $2^{15}-1$ 
(C) $2^{16}-1$ (D) $2^{15}-1$
19. What shell's wild-card is used to match any number of characters?
(A) [ijk] (B) ?
(C) * (D) [!ijk]
20. In OOP, which member of the superclass is never accessible to the subclass?
(A) Public member (B) Protected member
(C) Private member  (D) All of the members
21. What is encapsulation in object-oriented programming?
(A) The ability of a class to derive properties and behaviour from another class.
(B) The process of hiding the internal details of an object and providing a public interface to interact with it. 
(C) The mechanism that allows a class to have multiple methods with the same name but different parameters.
(D) The process of creating multiple instances of a class.
22. Which of the following data structures stores elements in a non-linear hierarchical order?
(A) Array (B) Stack
(C) Queue (D) Tree 
23. Which data structure is used to implement depth-first search (DFS) algorithm?
(A) Stack 
(B) Queue
(C) Linked list  (D) Tree
24. Cache memory works on the principle of
(A) Locality of data (B) Locality of memory
(C) Locality of reference  (D) Locality of reference and memory 

25. Which of the following is correct:
- (A) T flip flop can be made from J-K flip flop by connecting J and K terminals
 - (B) D - Flip flop can be constructed from J-K flip flop by connecting J terminal with K terminal through a NOT gate.
 - (C) Both (A) and (B) is correct.
 - (D) None of the above is correct.
26. The address resolution protocol (ARP) is used for:
- (A) Finding the IP address from the Domain name server (DNS)
 - (B) Finding the MAC address from the Domain name server (DNS) 
 - (C) Finding the IP address that corresponds to a MAC address
 - (D) Finding the MAC address that corresponds to an IP address
27. 129. 168.10.10 is a _____ IP address.
- (A) class A
 - (B) class B 
 - (C) class C
 - (D) class D
28. Which of the following is an application layer protocol?
- (A) UDP
 - (B) TCP
 - (C) HTTP 
 - (D) ARP
29. Which of the following is a 128-bit address?
- (A) MAC address
 - (B) IPv4 address 
 - (C) IPv6 address
 - (D) Ethernet address
30. Which of the following layer of the OSI model transforms a raw transmission facility into a line that appears free of undetected transmission errors?
- (A) Physical Layer
 - (B) Data Link Layer 
 - (C) Network Layer
 - (D) Transport Layer

31. A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/hr, what is the length of the platform?

(A) 120 m

(B) 240 m

(C) 300 m

(D) 400 m

32. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

(A) 12 days

(B) 15 days

(C) 16 days

(D) 18 days

33. Ayesha's father was 38 years of age when she was born, while her mother was 36 years old when her brother four years younger to her was born. What is the difference between the ages of her parents?

(A) 2 years

(B) 4 years

(C) 6 years

(D) 8 years

34. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random without replacement. What is the probability that none of the balls drawn is blue?

(A) 10/21

(B) 11/21

(C) 2/7

(D) 5/7

35. Look at this series: 36, 34, 30, 28, 24, ... What number should come next?

(A) 20

(B) 22

(C) 23

(D) 26

36. CMM, EOO, GQQ, _____, KUU

(A) GRR

(B) GSS

(C) ISS

(D) ITT

37. Consider the following statements.

Statement 1: Jessica has four children.

Statement 2: Two of the children have blue eyes and two of the children have brown eyes.

Statement 3: Half of the children are girls.

If the above three statements are facts, which of the following statements must also be a fact?

I: At least one girl has blue eyes.

II: Two of the children are boys.

III: The boys have brown eyes.

(A) I only

(B) II only

(C) II and III only

(D) None of the statements is a known fact.

38. Choose the word which is different from the rest.

(A) Sparrow

(B) Swan

(C) Parrot

(D) Koel

39. In a group of 100 people, 52 like mango and 68 like apple, and each person likes at least one of the two fruits. How many like both mango and apple?

(A) 52

(B) 48

(C) 20

(D) 68

40. $A - (B \cup C)$ is equal to

(A) $(A - B) \cup (A - C)$

(B) $A - B - C$

(C) $(A - B) \cap (A - C)$

(D) $A - (B \cap C)'$

[where symbol " $'$ " indicate complement]

41. How many different ways can the letters of the word "BANANA" be arranged?

(A) 720

(B) 60

(C) 120

(D) 360

42. Evaluate the composite function $f(g(x))$, given $f(x) = 3x + 1$ and $g(x) = 2x - 4$.
 (A) $6x - 2$ (B) $6x + 2$
 (C) $6x - 11$ (D) $6x + 11$
43. Which of the following statements is true about the empty set (ϕ)?
 (A) It is a subset of every set ✓
 (B) It is a superset of every set
 (C) It is an element of every set
 (D) It is equal to the universal set explain
44. A jar contains 8 red balls and 5 green balls. If two balls are drawn without replacement, what is the probability of drawing two red balls?
 (A) $4/39$ (B) $8/39$
 (C) $8/13$ (D) $14/39$
45. In how many ways can a group of 4 people be selected from a group of 10 people if two specific individuals must be included?
 (A) 36 (B) 28
 (C) 84 (D) 210
46. Which of the following relations is the reflexive relation over the set $\{1, 2, 3, 4\}$?
 (A) $\{(0,0), (1,1), (2,2), (2,3)\}$ (B) $\{(1,1), (1,2), (2,2), (3,3), (4,3), (4,4)\}$
 (C) $\{(1,1), (1,2), (2,1), (2,3), (3,4)\}$ ✗ (D) $\{(0,1), (1,1), (2,3), (2,2), (3,4), (3,1)\}$
47. If X_n be the set with n elements, where n is any finite cardinal number, then the cardinal number of its power set is -
 (A) 2^{n-1} ✗ (B) 2^n
 (C) 2^{n+1} (D) None of these
48. The sum of 40 terms of an A.P., whose first term is 2 and common difference is 4, will be
 (A) 3200 ✓ (B) 1600
 (C) 200 (D) 2800

49. What is the 2's complement representation of the binary number 01100110 in an 8-bit system?
- (A) 11111111 (B) 10011001
(C) 10011010 (D) 10011011

50. The Value of $\sqrt{42 + \sqrt{42 + \sqrt{42 + \sqrt{42 + \sqrt{42 + \sqrt{42 + \sqrt{42 + \dots \infty}}}}}}}$ is
- (A) $7/2, 4$ (B) $14, -3$
(C) $7, -6$ (D) $21, 2$

51. The value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \cos 4^\circ \cos 5^\circ \cos 6^\circ \cos 7^\circ \dots \cos 90^\circ$ is
- (A) 0 (B) 1
(C) $1/2$ (D) None of these

52. Find the domain of the function $f(x) = \sqrt{-4 - 3x + x^2}$
- (A) $[-1, 4]$ (B) $[2, -1]$
(C) $(-\infty, 0)$ (D) $(-\infty, \infty)$

53. The equation of straight line parallel to X-axis and pass through $(6, -7)$ is
- (A) $x = 6$ (B) $x = -7$
(C) $y = 6$ (D) $y = -7$

54. Assume for an unknown number system with base x following expression is valid $(122)_x = (37)_{10}$. Find out the value of base x.
- (A) 5 (B) 7
(C) 8 (D) 2

55. If, $T(n) = T(n/2) + 1$ and $T(1) = 1$, then $T(n) =$
- (A) $O(\log_2 n)$ (B) $O(2 \log_2 n)$
(C) $O(n \log_2 n)$ (D) $O(n^2)$
- Learn time and space complexity

56. The time complexity of the Quick sort algorithms in the worst case is
(A) $O(n \log_2 n)$  (B) $O(n^2 \log_2 n)$
(C) $O(\log_2 n)$ (D) $O(n^2)$
57. Eight Queen Problem can be solved by.
(A) Divide and conquer Strategy (B) Backtracking 
(C) Greedy Strategy (D) Sorting
58. Which one of the following Algorithm solves all pairshortest paths?
(A) Krushal's (B) Prim's 
(C) Dijkstra's (D) Floyd-Warshall
59. Which of the following data structure finds its use in recursion?
(A) Stack (B) Queue
(C) Binary Tree (D) Linked List 
60. A balance factor in AVL tree is used to check
(A) what rotation to make. (B) if all child nodes are at same level. 
(C) when the last rotation occurred. (D) if the tree is unbalanced.

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